CASE STUDY 3: ATM Transaction

Python Activity-Based Learning

1. Problem Statement

Write a simple Python program to simulate an ATM transaction. Start with a balance of ■5000, accept a deposit and a withdrawal, update the balance, and display the final balance.

2. Objectives

• Initialize an account balance.

• Add a deposit to the balance.

• Check and process a withdrawal.

3. Assumptions

• The starting balance is ■5000.

• Deposit and withdrawal amounts cannot be negative.

• A withdrawal cannot be greater than the available balance.

4. Algorithm

1. Start.

2. Set balance = 5000.

3. Read deposit amount and add it to balance if valid.

4. Read withdrawal amount.

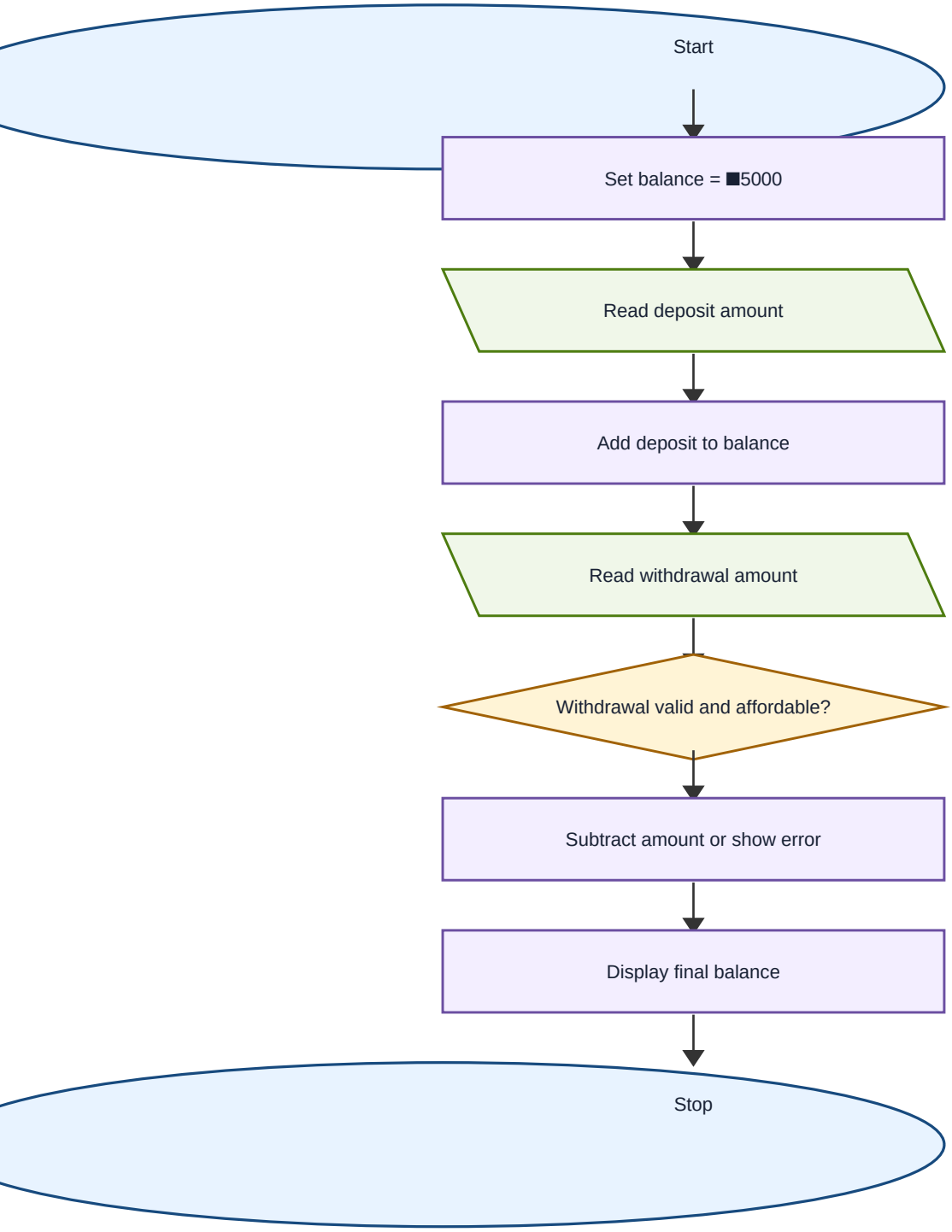
5. If withdrawal is valid and does not exceed balance, subtract it; otherwise show an error.

6. Display final balance.

7. Stop.

5. Flowchart

Oval = Start/Stop; parallelogram = Input; rectangle = Process; diamond = Decision.



6. Pseudocode

```
BEGIN
  balance = 5000
  INPUT deposit
  IF deposit >= 0 THEN
      balance = balance + deposit
  END IF
  INPUT withdrawal
  IF withdrawal >= 0 AND withdrawal <= balance THEN
      balance = balance - withdrawal
  ELSE
      PRINT "Invalid withdrawal"
  END IF
  PRINT balance
END
```

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7. Python Source Code

```
balance = 5000
print("Opening balance: ■", balance)

deposit = float(input("Enter deposit amount: "))
if deposit >= 0:
    balance = balance + deposit
else:
    print("Invalid deposit amount")

withdrawal = float(input("Enter withdrawal amount: "))
if withdrawal < 0:
    print("Invalid withdrawal amount")
elif withdrawal <= balance:
    balance = balance - withdrawal
else:
    print("Insufficient balance")

print("Final balance: ■", balance)
```

8. Test Cases

Input	Expected Output
Deposit 1000; withdraw 500	Final balance: ■5500.0
Deposit 0; withdraw 6000	Insufficient balance; Final balance: ■5000

9. Sample Output

Run the Python program and enter the values from one of the test cases above. The displayed result should match the expected output.

10. Conclusion

The program demonstrates how an ATM updates a balance using input, arithmetic operations and conditional checks.